

Whisper Fine-Tuning Configuration

Training Configuration

All the hyperparameters besides the Learning Rate were the same for the three sizes of Whisper (Tiny, Small, and Whisper Large-V3)

Hyperparameter Configuration

Model-Specific Settings

Whisper Tiny

- Learning Rate: 5×10^{-5}
- Per-device Batch Size: 256
- Attention Implementation: Scaled Dot-Product Attention

Whisper Small

- Learning Rate: 1×10^{-5}
- Per-device Batch Size: 256
- Attention Implementation: Scaled Dot-Product Attention

Whisper Large V3

- Learning Rate: 5×10^{-6}
- Per-device Batch Size: 256
- Attention Implementation: Scaled Dot-Product Attention

Global Training Parameters

- Optimizer: AdamW with fused implementation
- Mixed Precision: BF16
- Warmup Ratio: 0.1
- Gradient Checkpointing: Enabled
- Maximum Epochs: 5

Data Configuration

- Audio Sampling Rate: 16,000 Hz
- Languages: Portuguese (CAPES dataset) and Dutch
- Training Data: Quality-filtered synthetic audio and the original Common Voice 17
- Validation/Test Data: Common Voice validation and test sets.
- Data Loader Workers: 30 workers